Test Series No.

# 1.13

# MASTERY TEST SERIES

**DURATION: 180 Minutes** | **M. MARKS: 720**

**DATE: 21/03/2026**

## ANSWER KEY

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. | (4) | 46. | (1) | 91. | (2) | 136. | (2) |
| 2. | (3) | 47. | (1) | 92. | (1) | 137. | (1) |
| 3. | (2) | 48. | (2) | 93. | (4) | 138. | (4) |
| 4. | (4) | 49. | (1) | 94. | (3) | 139. | (2) |
| 5. | (4) | 50. | (1) | 95. | (1) | 140. | (1) |
| 6. | (1) | 51. | (3) | 96. | (1) | 141. | (3) |
| 7. | (1) | 52. | (3) | 97. | (2) | 142. | (2) |
| 8. | (3) | 53. | (3) | 98. | (4) | 143. | (2) |
| 9. | (2) | 54. | (3) | 99. | (2) | 144. | (3) |
| 10. | (3) | 55. | (4) | 100. | (4) | 145. | (1) |
| 11. | (4) | 56. | (1) | 101. | (1) | 146. | (2) |
| 12. | (2) | 57. | (4) | 102. | (3) | 147. | (4) |
| 13. | (1) | 58. | (4) | 103. | (2) | 148. | (1) |
| 14. | (4) | 59. | (4) | 104. | (2) | 149. | (3) |
| 15. | (1) | 60. | (3) | 105. | (3) | 150. | (3) |
| 16. | (3) | 61. | (2) | 106. | (4) | 151. | (3) |
| 17. | (4) | 62. | (4) | 107. | (1) | 152. | (1) |
| 18. | (2) | 63. | (4) | 108. | (3) | 153. | (2) |
| 19. | (3) | 64. | (1) | 109. | (2) | 154. | (3) |
| 20. | (4) | 65. | (4) | 110. | (1) | 155. | (1) |
| 21. | (2) | 66. | (3) | 111. | (2) | 156. | (3) |
| 22. | (2) | 67. | (1) | 112. | (3) | 157. | (3) |
| 23. | (1) | 68. | (4) | 113. | (3) | 158. | (3) |
| 24. | (4) | 69. | (4) | 114. | (3) | 159. | (2) |
| 25. | (3) | 70. | (4) | 115. | (2) | 160. | (1) |
| 26. | (3) | 71. | (3) | 116. | (2) | 161. | (2) |
| 27. | (4) | 72. | (1) | 117. | (4) | 162. | (3) |
| 28. | (2) | 73. | (3) | 118. | (2) | 163. | (3) |
| 29. | (1) | 74. | (2) | 119. | (1) | 164. | (2) |
| 30. | (3) | 75. | (3) | 120. | (3) | 165. | (2) |
| 31. | (2) | 76. | (2) | 121. | (2) | 166. | (3) |
| 32. | (4) | 77. | (4) | 122. | (4) | 167. | (1) |
| 33. | (4) | 78. | (4) | 123. | (1) | 168. | (3) |
| 34. | (3) | 79. | (1) | 124. | (3) | 169. | (1) |
| 35. | (3) | 80. | (4) | 125. | (2) | 170. | (3) |
| 36. | (2) | 81. | (2) | 126. | (4) | 171. | (4) |
| 37. | (1) | 82. | (4) | 127. | (4) | 172. | (1) |
| 38. | (4) | 83. | (3) | 128. | (3) | 173. | (3) |
| 39. | (2) | 84. | (3) | 129. | (3) | 174. | (4) |
| 40. | (3) | 85. | (2) | 130. | (1) | 175. | (2) |
| 41. | (3) | 86. | (4) | 131. | (3) | 176. | (3) |
| 42. | (3) | 87. | (1) | 132. | (4) | 177. | (1) |
| 43. | (2) | 88. | (3) | 133. | (2) | 178. | (3) |
| 44. | (1) | 89. | (1) | 134. | (4) | 179. | (4) |
| 45. | (4) | 90. | (1) | 135. | (1) | 180. | (2) |



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# Hints & Solutions

**1. (4)**

Work ($W$): $[ML^2T^{-2}]$

Charge ($q$): $[AT]$ (since Charge = Current × Time)

$$[\text{emf}] = \frac{[ML^2T^{-2}]}{[AT]} = [ML^2T^{-3}A^{-1}]$$

**(NCERT Ed. 2025-26, 12th, Page No. 94)**

**2. (3)**

Equivalent spring constant,

$$k_{eq} = \frac{\left(\frac{k}{3}\right)\left(\frac{2k}{3}\right)}{\frac{k}{3}+\frac{2k}{3}} = \frac{2k}{9}$$

Time period, $T = 2\pi\sqrt{\frac{m}{k_{eq}}} = 2\pi\sqrt{\frac{9m}{2k}} = 6\pi\sqrt{\frac{m}{2k}}$

**(NCERT Ed. 2025-26, 11th, Page No. 268)**

**3. (2)**

$$\vec{B}_P = \vec{B}_A + \vec{B}_{AB} + \vec{B}_B$$

$$\Rightarrow \vec{B}_P = \frac{\mu_0 i}{4\pi R}(\otimes) + \frac{\mu_0 i}{4R}(\cdot) + \frac{\mu_0 i}{4\pi R}(\otimes)$$

$$\Rightarrow \vec{B}_p = \frac{\mu_0 i}{4R}\left(1-\frac{2}{\pi}\right) \text{ away from the page}$$

**(NCERT Ed. 2025-26, 12th, Page No. 116)**

**4. (4)**

If $\Phi$ total $= \int E \cdot dS = 0$, it means the net electric flux through the surface is zero. This happens when the net charge enclosed is zero, OR more generally, when the number of electric field lines entering the surface equals the number of field lines leaving it.

**(NCERT Ed. 2025-26, 12th, Page No. 30)**

**5. (4)**

$$\frac{\lambda_p}{\lambda_e} = \frac{\frac{h}{m_p v_p}}{\frac{h}{m_e v_e}} = \frac{m_e v_e}{m_p v_p}$$

$$1.813\times10^{-4} = \frac{9.1\times10^{-31}}{m_p}\times\frac{1}{3}$$

$m_p = 1.675\times10^{-27}$ kg (Neutron)

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**6. (1)**

Reactances & Impedance:

$$X_L = \omega L = 100\times0.1 = 10\,\Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{100\times500\times10^{-6}} = 20\,\Omega$$

$$Z = \sqrt{R^2+(X_C-X_L)^2} = \sqrt{10^2+10^2} = 10\sqrt{2}\,\Omega$$

Current ($I$):

$$I_0 = \frac{V_0}{Z} = \frac{200}{10\sqrt{2}} = 10\sqrt{2}\text{ A}$$

Phase difference

$$\phi = \tan^{-1}\left(\frac{X_C - X_L}{R}\right) = \tan^{-1}(1) = +\frac{\pi}{4}$$

(Current leads $V$)

$$\Rightarrow I(t) = 10\sqrt{2}\sin\left(100t+\frac{\pi}{4}\right)$$

Capacitor Voltage ($V_C$):

Peak voltage

$$(V_C)_0 = I_0 X_C = (10\sqrt{2})(20) = 200\sqrt{2}\text{ V}$$

Since $V_C$ lags current by $\frac{\pi}{2}$, its phase is

$$\left(\frac{\pi}{4}-\frac{\pi}{2}\right) = -\frac{\pi}{4}$$

$$\Rightarrow V_C(t) = 200\sqrt{2}\sin\left(100t-\frac{\pi}{4}\right)$$

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**7. (1)**

$$\vec{\tau} = \vec{r}\times\vec{F}$$

$$\vec{\tau} = (\hat{i}+\hat{j})\times(5\sqrt{3}\hat{i}+5\hat{j})$$

$$\vec{\tau} = 5(\hat{i}\times\hat{j}) + 5\sqrt{3}(\hat{j}\times\hat{i})$$

$$\vec{\tau} = 5\hat{k} - 5\sqrt{3}\hat{k}$$

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**8. (3)**

Total Mass ($M$):

$$M = 2m+3m+2m+m = 8m$$

Total Moment of Inertia ($I$) about the central perpendicular axis:

Since all masses are at a distance '$a$' from the center:

$$I = (2m)a^2 + (3m)a^2 + (2m)a^2 + (m)a^2$$

$$I = 8ma^2$$

Radius of Gyration ($k$): $I = Mk^2$

$$8ma^2 = (8m)k^2$$

$$k^2 = a^2 \Rightarrow k = a$$

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**9. (2)**

Candela is the unit of luminous intensity.

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**10. (3)**

$F_{\text{parallel}} = F\sin\theta = 40\times\sin(30°) = 40\times 0.5 = 20\text{ N}$

$\text{Shear Stress} = \dfrac{\text{Parallel Force}}{\text{Area}}$

$\tau = \dfrac{20\text{ N}}{2\times10^{-4}\text{ m}^2}$

$\tau = 10\times10^4\text{ N/m}^2 = 1.0\times10^5\text{ N/m}^2$

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**11. (4)**

Statement 1 is Correct: Reversibility and quasi-static (infinitely slow) movement are essential to ensure no energy is lost to entropy or friction.

Statement 2 is Incorrect: The piston must move infinitely slowly (quasi-statically) to maintain equilibrium. Rapid movement would cause turbulence and irreversibility.

Statement 3 is Correct: Infinite thermal capacity ensures that the temperatures ( $T_H$ and $T_L$ ) do not change regardless of how much heat is transferred.

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**12. (2)**

Total Reading = MSR + (VSC × LC)

$3.6\text{ cm} + (4\times0.01\text{ cm}) = 3.64\text{ cm}$

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**13. (1)**

Relationship: For specific heat per unit mass, the formula is $C_p - C_v = \dfrac{R}{M}$.

Molar Mass ( $M$ ): For Nitrogen gas ( $N_2$ ), $M = 28\text{ g/mol}$.

Result: Substituting $M$, we get $C_p - C_v = \dfrac{R}{28}$.

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**14. (4)**

$E$ due to ring at axis $= \dfrac{1}{4\pi\varepsilon_0}\dfrac{Qx}{(x^2+R^2)^{3/2}}$

here $Q = 4\times10^{-9}$ C

$R = 0.3$ m

$x = 0.4$ m

$E = 115.2$ V/m

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**15. (1)**

In option (1), P is at lower potential and N is at higher potential, So it is reversed biased.

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**16. (3)**

Under a constant applied voltage, the electric field ( $E$ ) in the wire is constant. We know the relation between current density, conductivity, and drift velocity is:

$J = \sigma E = nev_d$

$\Rightarrow v_d \propto \sigma$

As temperature increases, increased lattice vibrations cause the relaxation time ( $\tau$ ) to decrease, which decreases the conductivity ( $\sigma$ ). Because $v_d \propto \sigma$, the drift velocity ( $v_d$ ) also decreases.

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**17. (4)**

$F\times s = \dfrac{1}{2}mv^2 - 0$

$(2\times10^4)\times3 = \dfrac{1}{2}(5\times10^7)\times v^2$

$6\times10^4 = 2.5\times10^7\times v^2$

$v^2 = \dfrac{6\times10^4}{2.5\times10^7} = 24\times10^{-4}$

$v = \sqrt{24\times10^{-2}} \approx 0.05\text{ m/s}$

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**18. (2)**

$Y_1\alpha_1\Delta\theta = Y_2\alpha_2\Delta\theta$

$\alpha_2 = \dfrac{1.2\times10^{11}\times1.5\times10^{-5}}{1.5\times10^{11}}$

$\alpha_2 = 1.2\times10^{-5}/°\text{C}$

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**19. (3)**

Let junction potential is $V$ –

By using kirchoff's first law –

$\dfrac{24-V}{3} + \dfrac{10-V}{2} + \dfrac{9-V}{1} = 0$

$V = 12V$

$I = \dfrac{24-12}{3} = 4A$

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**20. (4)**

$p_x = 2t^2$ $\qquad$ $p_y = 3t$

$F_x = \dfrac{dp_x}{dt}$ $\qquad$ $F_y = \dfrac{dp_y}{dt}$

$F_x = 4t$ $\qquad$ $F_y = 3$

At $t = 1$ s, $F_x = 4$ N, $F_y = 3$ N

$F_{net} = \sqrt{F_x^2 + F_y^2}$

$= \sqrt{4^2+3^2} = 5\text{ N}$

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**21. (2)**

So Energy in electric field and magnetic field are same.

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**22. (2)**

Using the formula for Potential Energy, $U = -MB\cos\theta$:

(A) $\theta = 0° \Rightarrow U = -MB(1) = -MB \rightarrow (IV)$

(B) $\theta = 180° \Rightarrow U = -MB(-1) = +MB \rightarrow (II)$

(C) $\theta = 90° \Rightarrow U = -MB(0) = 0 \rightarrow (I)$

(D) $\theta = 60° \Rightarrow U = -MB(1/2) = \frac{-MB}{2} \rightarrow (III)$

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**23. (1)**

$v^2 = u^2 + 2as$

$\Rightarrow v^2 = (0.4)^2 + 2\times10\times0.4$

$= 0.16 + 8$

$\Rightarrow v = \sqrt{8.16}$ m/s

$\Rightarrow v \approx 2.9$ m/s

Now, $A_1v_1 = A_2v_2$

$\Rightarrow \frac{\pi d_1^2 v_1}{4} = \frac{\pi d_2^2 v_2}{4}$

$\Rightarrow d_1^2 v_1 = d_2^2 v_2$

$\Rightarrow (8\times10^{-3})^2 \times 0.4 = d_2^2 \times 2.9$

$\Rightarrow d_2^2 = \frac{64\times4}{29}\times10^{-6}$

$d_2 \approx 2.9\times10^{-3}$ m

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**24. (4)**

Power loss is minimum when resistance is low, inductance is high.

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**25. (3)**

Force produced by the engine

$$F = \frac{P}{v} = \frac{30\times10^3}{30} = 10^3 \text{ N}$$

$$\text{Acceleration} = \frac{(\text{Forward force by engine}) - (\text{resistive force})}{\text{mass of car}}$$

$$\frac{1000-750}{1250} = \frac{250}{1250} = \frac{1}{5} \text{ m/s}^2$$

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**26. (3)**

Wave transports both energy and momentum. If the temperature is fixed speed of sound in air is independent of pressure.

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**27. (4)**

$Ke_{max} = hv - hv_0$

$\Rightarrow v_0 = v - \frac{Ke_{max}}{h} = 10^{15} - \frac{3\times1.6\times10^{-19}}{6.63\times10^{-34}}$

$= 2.7\times10^{14}$ Hz $\approx 3\times10^{14}$ Hz

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**28. (2)**

Gravitational force on earths surface is $F = \frac{GM_em}{R_e^2}$

And gravitational force at height 1600 km from earths surface is $\frac{GM_em}{\left(\frac{5R_e}{4}\right)^2} = \frac{16F}{25}$

$\therefore$ percentage loss $= \frac{9}{25}\times100 = 36\%$

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**29. (1)**

$\phi = B.A$

$= at.\pi r^2$

$\varepsilon = \frac{-d\phi}{dt} = -\pi r^2 a$

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**30. (3)**

LEDs(Light Emitting Diodes) which are usually semiconductors that have the ability to convert electrical energy to light energy. The color of the emitted light totally depends on the type of semiconductor material that has been used in it. Depending on the LEDs emitting region can be classified into their wavelengths: UV, Visible and Infrared.

**31. (2)**

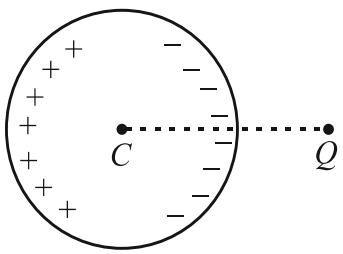
$9000 = \frac{KA\Delta T}{l}$

$\Rightarrow \Delta T = \frac{9000\times1}{200\times0.75}$

$\Rightarrow \Delta T = 60°C$

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**32. (4)**




Potential at $C$ due to induced charge is zero.

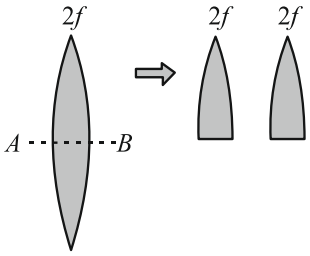
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**33. (4)**

If an equiconvex lens of focal length $2f$ is cut into two identical parts by a plane along the optical axis, the focal length of each part will be equal to that of the initial lens; because $\mu$, $R_1$ and $R_2$ will remain unchanged. Only intensity of image will be reduced.




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**34. (3)**

$a = -a_0 \sin \omega t = -\omega^2 x$

$x = \frac{a_0}{\omega^2} \sin \omega t$

**(NCERT Ed. 2025-26, 11th, Page No. 262)**

**35. (3)**

A ball is thrown under gravity from ground, crosses at time $t_1$ and $t_2$ at height $h$ from the ground, then sum of these times is equal to its total flight time.

$T = t_1 + t_2 = 4 + 12 = 16$ s

$T = \frac{2u}{g} \therefore u = \frac{gT}{2} = \frac{10 \times 16}{2} = 80$ m/s

**(NCERT Ed. 2025-26, 11th, Page No. 20)**

**36. (2)**

$\text{emf} = -\frac{d\phi}{dt}$

$0 \rightarrow t_1$; $|e| =$ constant

$t_1 \rightarrow t_2$ ; $e = 0$

$t_2 \rightarrow$ end ; $|e|$ = constant

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**37. (1)**

At constant pressure, heat given is $nc_p\Delta T$ and work done is $nR\Delta T$

$\therefore \frac{nR\Delta T}{nC_p\Delta T} = \frac{R}{\frac{\gamma R}{\gamma - 1}} = \frac{\gamma - 1}{\gamma} = \frac{2/3}{5/3} = 40\%$

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**38. (4)**

$\vec{L} = m(\vec{r} \times \vec{v})$

$= 2(\hat{i} + \hat{j}) \times (2\hat{i} - 2\hat{j} + 2\hat{k})$

$= 4\hat{i} - 4\hat{j} - 8\hat{k}$

Angular momentum about z axis is $-8\hat{k}$

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**39. (2)**

$V_n = 2.18 \times 10^6 \times \frac{1}{n}$

$V_2 = \frac{2.18 \times 10^6}{2} = 1.09 \times 10^6$ m/s

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**40. (3)**

Binding Energy of $A = 240 \times 7.6 = 1824$ MeV

Binding Energy of $B = 100 \times 8.1 = 810$ MeV

Binding Energy of $C = 140 \times 8.1 = 1134$ MeV

So

$Q = (810 + 1134)$ MeV $- 1824$ MeV $= 120$ MeV

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**41. (3)**

Spread of diffraction region will be decreased and mid band becomes narrow.

**(NCERT Ed. 2025-26, 12th, Page No. 267)**

**42. (3)**

In case of Fraunhofer diffraction the incident wavefront is planar.

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**43. (2)**

Three different materials are used in the lens and the focal length of the lens depends on the material of the lens. Hence, three different images will be formed as the ray will converge at three different points.

**(NCERT Ed. 2025-26, 12th, Page No. 237)**

**44. (1)**

From Ampere's circular law.

$\oint \vec{B}.d\vec{l} = \mu_0 i_{in}$

Where $\vec{B}$ is net magnetic field produced due to all steady currents, So $\vec{B} = \vec{B}_a + \vec{B}_b + \vec{B}_c + \vec{B}_d$

**(NCERT Ed. 2025-26, 12th, Page No. 118)**

**45. (4)**

On inserting dielectric $C$ increases. As battery is connected $V$ is constant.

From $Q = CV$, $Q$ increases $\Rightarrow$ $A$ is correct.

Energy $U = \frac{1}{2}CV^2$, as $V$ is constant and $C$ increase

$\therefore$ $U$ increases. $\Rightarrow$ $B$ is correct.

As charge increases on the capacitor, some work is done by battery in moving the positive charge from negative plate to the positive plate.

$\Rightarrow$ $C$ is correct.

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**46. (1)**

- Endothermic reactions on increasing temperature shifts in forward direction.
- On increasing pressure, reaction move in that direction where number of gaseous moles decreases.

**(NCERT Ed. 2025-26 11th Part-I Page No. 187)**

**47. (1)**

Basic nature: I > III > II > IV

**(NCERT Ed. 2025-26 12th Part-II Page No. 269)**

**48. (2)**

$w = -\int P_{ext}\, dV = 0$

$\Delta U = nC_v\Delta T = 0$

$\therefore\ q + w = \Delta U \qquad \therefore\ \ q = 0$

**(NCERT Ed. 2025-26 11th Part-I Page No. 142)**

**49. (1)**

- $\overset{-2}{O}=\overset{+2}{C}=\overset{+2}{C}=\overset{+2}{C}=\overset{-2}{O}$

  $3x + 2(-2) = 0$

  $x = +\frac{4}{3}$

- $Fe_3O_4 : FeO + Fe_2O_3$

**(NCERT Ed. 2025-26 11th Part-II Page No. 245)**

**50. (1)**

Diagonal relationship:

Li → Mg, Be → Al, B → Si

**(NCERT Ed. 2025-26 11th Part-I Page No. 93)**

**51. (3)**

Kjeldahl method is not applicable to estimate nitrogen in nitrobenzene because nitrogen present in nitrobenzene does not change to $(NH_4)_2SO_4$ during the process.

**(NCERT Ed. 2025-26 11th Part-II Page No. 288)**

**52. (3)**

In $O_2F_2$, oxidation state of oxygen is positive because fluorine is more electronegative than oxygen.

**(NCERT Ed. 2025-26 11th Part-II Page No. 240)**

**53. (3)**

Dalton's theory could not explain the laws of gaseous volumes.

**(NCERT Ed. 2025-26 11th Part-I Page No. 16)**

**54. (3)**

Since graph touches at one point along radial distance axis so it has one radial node and at r = 0, it has some value so it should be a s-orbital ($l = 0$).

$\therefore\ n - \ell - 1 = 1$

$n - 0 - 1 = 1$

$n = 2$

so, 2s-obital.

**(NCERT Ed. 2025-26 11th Part-I Page No. 59)**

**55. (4)**

All the given statements are shortcomings of valence bond theory for coordination compounds.

**(NCERT Ed. 2025-26 12th Part-I Page No. 131)**

**56. (1)**

Among 3d series, Mn shows maximum number of oxidation states.

**(NCERT Ed. 2025-26 12th Part-I Page No. 96)**

**57. (4)**

Unit of $k = \left(\frac{mol}{lit}\right)^{1-n} time^{-1}$

**(NCERT Ed. 2025-26 12th Part-I Page No. 69)**

**58. (4)**

$Gypsum : CaSO_4 \cdot 2H_2O$

**59. (4)**

- Vapour pressure of water increases on increasing the temperature.
- Vapour pressure is not a colligative property.

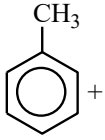
**(NCERT Ed. 2025-26 12th Part-I Page No. 09)**

**60. (3)**

- Cyclic polymerization:

  $3C_2H_2 \xrightarrow[\text{Tube, 873 K}]{\text{Red hot iron}} C_6H_6$

- Aromatization:

  n-Heptane $\xrightarrow[\text{773 K, 10-20 atm}]{Cr_2O_3}$  $+ 4H_2$

- Dehalogenation: $ClCH_2CH_2Cl \xrightarrow[\Delta]{Zn} C_2H_4$

- Isomerisation:

  n-Butane $\xrightarrow{\text{Anhyd. } AlCl_3 + HCl}$ $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_3$

**(NCERT Ed. 2025-26 11th Part-II Page No. 301, 304, 309)**



[Contd...

**61. (2)**

- $HNO_3 : [H^+] = 10^{-3}$ M

  $pH = -\log(10^{-3}) = 3$

- $H_2SO_4 : [H^+] = 0.01 \times 2 = 2 \times 10^{-2}$ M

  $pH = -\log(2 \times 10^{-2}) = 1.7$

- $NaOH : [OH^-] = 10^{-7} + 10^{-7} = 2 \times 10^{-7}$ M

  $pOH = -\log(2 \times 10^{-7}) = 7 - 0.3$

  $pH = 14 - 6.7 = 7.3$

- $NaOH : [OH^-] = 10^{-2}$ M

  $pOH = -\log(10^{-2}) = 2$

  $pH = 14 - 2 = 12$

**(NCERT Ed. 2025-26 11th Part-I Page No. 194)**

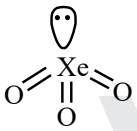
**62. (4)**

For higher concentration of $H_2SO_4$ at anode:

$2SO_4^{2-}(aq) \rightarrow S_2O_8^{2-}(aq) + 2e^-$

**(NCERT Ed. 2025-26 12th Part-I Page No. 54)**

**63. (4)**

- $CO_2 : O=C=O \leftrightarrow \overset{\oplus}{O}\equiv C-\overset{\ominus}{O} \leftrightarrow \overset{\ominus}{O}-C\equiv\overset{\oplus}{O}$
- $SO_4^{2-}$ : Bond order $= \dfrac{2+1+1+2}{4} = 1.5$
- $XeO_3$: 
- $CO_3^{2-}$ : Bond order $= \dfrac{2+1+1}{3} = 1.33$

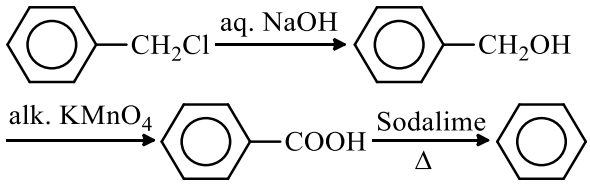
**(NCERT Ed. 2025-26 11th Part-I Page No. 129)**

**64. (1)**

- Fehling solution contains tartarate salt.
- Reagent used in Etard reaction is chromyl chloride.

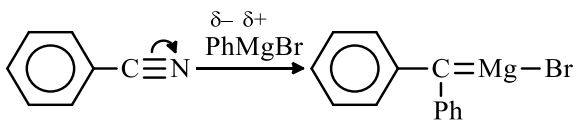
**(NCERT Ed. 2025-26 12th Part-II Page No. 239)**

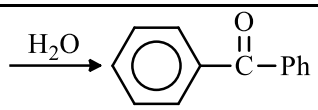
**65. (4)**




**(NCERT Ed. 2025-26 12th Part-II Page No. 210)**

**66. (3)**




Benzophenone

**(NCERT Ed. 2025-26 12th Part-II Page No. 234)**

**67. (1)**

Reactivity of alcohols towards Lucas reagent:

3° > 2° > 1°

**(NCERT Ed. 2025-26 12th Part-II Page No. 208)**

**68. (4)**

Electron withdrawing group ($–NO_2$), increases the reactivity of Haloarenes towards $ArS_N$ reaction.

**(NCERT Ed. 2025-26 12th Part-II Page No. 182)**

**69. (4)**

More the stability of intermediate carbocation, more will be reactivity of alkyl halide towards $S_N1$ reaction.

**(NCERT Ed. 2025-26 12th Part-II Page No. 174)**

**70. (4)**

Acidic nature: 4 > 1 > 2 > 3

**(NCERT Ed. 2025-26 12th Part-II Page No. 251)**

**71. (3)**

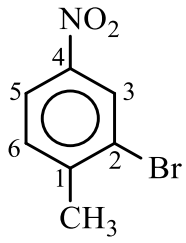
Carbon can show property of catenation because C–C bonds are very strong.

**72. (1)**

- Nitration: $E^+ = NO_2^+$
- Sulphonation: $E^+ = SO_3$

**(NCERT Ed. 2025-26 11th Part-II Page No. 322)**

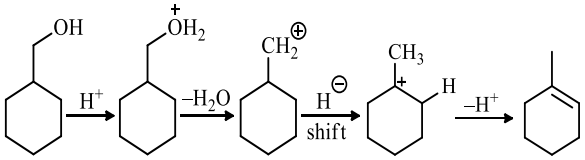
**73. (3)**




2-Bromo-1-methyl-4-nitrobenzene

**(NCERT Ed. 2025-26 11th Part-II Page No. 269)**

**74. (2)**




**(NCERT Ed. 2025-26 12th Part-II Page No. 209)**



[Contd...

**75. (3)**
Ethane forms only one monosubstituted derivative because all the hydrogens in ethane are equivalent.
**(NCERT Ed. 2025-26 11th Part-II Page No. 302)**

**76. (2)**
Globular proteins are usually soluble in water.
**(NCERT Ed. 2025-26 12th Part-II Page No. 293)**

**77. (4)**
$$t = \frac{2.303}{k}\log\frac{[A_0]}{[A]} = \frac{2.303}{0.1}\log\left[\frac{200}{2}\right]$$
$$= 23.03 \times 2 = 46.06\ s$$
**(NCERT Ed. 2025-26 12th Part-I Page No. 73)**

**78. (4)**
$B_2 : (\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_y^2 = \pi 2p_z^0)$

- $BO = \frac{N_b - N_a}{2} = \frac{6-4}{2} = 1$
- No unpaired electron so diamagnetic

**(NCERT Ed. 2025-26 11th Part-I Page No. 131)**

**79. (1)**
$$C_6H_6 + \frac{15}{2}O_2 \rightarrow 6CO_2 + 3H_2O$$
Vol. of $O_2$ required $= \frac{15}{2} \times n_{C_6H_6} \times 22.4\ L$
$$= \frac{15}{2} \times \frac{7.8}{78} \times 22.4 = 16.8\ L$$
**(NCERT Ed. 2025-26 11th Part-I Page No. 20)**

**80. (4)**
Acid-base reactions (neutralization reactions) are non-redox reactions.
**(NCERT Ed. 2025-26 11th Part-II Page No. 236)**

**81. (2)**

- Unu: mendelevium
- Unq: Rutherfordium
- Unt: Lawrencium
- Uns: Bohrium

**(NCERT Ed. 2025-26 11th Part-I Page No. 80)**

**82. (4)**
$$\Delta H^o = BE_R - BE_P$$
$$= (433 + 192) - (2 \times 364) = -103\ kJ\ mol^{-1}$$
**(NCERT Ed. 2025-26 11th Part-I Page No. 154)**

**83. (3)**
$KMnO_4$ titrations in presence of HCl are unsatisfactory because $KMnO_4$ oxidises HCl into $Cl_2$.
**(NCERT Ed. 2025-26 12th Part-I Page No. 108)**

**84. (3)**
$[Co(NH_3)_4Cl_2]^+$ is a heteroleptic coordination entity.
**(NCERT Ed. 2025-26 12th Part-I Page No. 122)**

**85. (2)**
Energy required $= 13.6\frac{Z^2}{n^2}eV$
$$= 13.6 \times \frac{2^2}{1^2}eV = 54.4\ eV$$
**(NCERT Ed. 2025-26 11th Part-I Page No. 48)**

**86. (4)**
$$2CH_3COOH \xrightarrow{C_6H_6} (CH_3COOH)_2$$
$(i)$ 1 mol 0
(f) $(1-0.8)$ mol $\left(\frac{0.8}{2}\right)$ mol

$$i = \frac{n_{after}}{n_{before}} = \frac{(1-0.8) + \left(\frac{0.8}{2}\right)}{1}$$
$\therefore\ i = 0.6$
**(NCERT Ed. 2025-26 12th Part-I Page No. 24)**

**87. (1)**
$$\Lambda_m = \frac{\kappa \times 1000}{M} = \frac{\left(\frac{1}{1\times10^{-4}}\right)\times 1000}{0.1}$$
$$= 1 \times 10^8\ ohm^{-1}\ cm^2\ mol^{-1}$$
**(NCERT Ed. 2025-26 12th Part-I Page No. 47)**

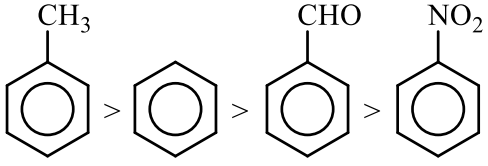
**88. (3)**
Borax bead test gives characteristic colours of transition metals.

**89. (1)**
Electronegativity on pauling scale:

| Al | Ga | In | Tl |
|---|---|---|---|
| 1.5 | 1.6 | 1.7 | 1.8 |

**(NCERT Ed. 2025-26 11th Part-I Page No. 91)**

**90. (1)**
Reactivity order:




**(NCERT Ed. 2025-26 12th Part-II Page No. 324)**


[Contd...

**91. (2)**

Watson met Crick and discovered their common interest in solving the DNA structure. Their first serious effort, was unsatisfactory. Their second effort based upon more experimental evidence and better appreciation of the nucleic acid literature, resulted, early in March 1953, in the proposal of the complementary double-helical configuration.

**(NCERT Ed. 2025-26, 12th, Page No. 80)**

**92. (1)**

$G_1$ phase corresponds to the interval between mitosis and initiation of DNA replication. It is follow by S or synthesis phase that marks the period during which DNA synthesis or replication takes place. After it comes the $G_2$ phase in which proteins are synthesised. It is followed by the M Phase that represents the phase when the actual cell division or mitosis occurs.

**(NCERT Ed. 2025-26, 11th, Page No. 121)**

**93. (4)**

The two-kingdom classification system did not distinguish between the eukaryotes and prokaryotes, unicellular and multicellular organisms. It also grouped together the unicellular organisms and the multicellular ones, say, for example, *Chlamydomonas* and *Spirogyra* were placed together under algae.

**(NCERT Ed. 2025-26, 11th, Page No. 11)**

**94. (3)**

While the membrane or lamellae of the grana have both PS I and PS II the stroma lamellae membranes lack PS II as well as NADP reductase enzyme. Since stroma lamellae contain Photosystem I, they possess the specific reaction centre (P700) and the light-harvesting complex (LHC I) associated with it.

**(NCERT Ed. 2025-26, 11th, Page No. 140)**

**95. (1)**

The root is covered at the apex by a thimble-like structure called the root cap. It protects the tender apex of the root as it makes its way through the soil. A few millimetres above the root cap is the region of meristematic activity. The cells of this region are very small, thin-walled and with dense protoplasm. They divide repeatedly. The cells proximal to this region undergo rapid elongation and enlargement and are responsible for the growth of the root in length. This region is called the region of elongation. The cells of the elongation zone gradually differentiate and mature. Hence, this zone, proximal to region of elongation, is called the region of maturation. From this region some of the epidermal cells form very fine and delicate, thread-like structures called root hairs.

**(NCERT Ed. 2025-26, 11th, Page No. 59)**

**96. (1)**

Man - $I^AI^A$

Woman - $I^BI^B$

When they cross, all offspring will inherit one $I^A$ allele from one parent and one $I^B$ allele from the other, resulting in the genotype $I^AI^B$ i.e. blood group AB. O blood group genotype is *ii* which is not possible in this situation. For blood group O, the child must inherit one *i* allele from each of the parents.

**(NCERT Ed. 2025-26, 12th, Page No. 61)**

**97. (2)**

Casparian strips are water-impermeable waxy deposits of suberin found in the endodermis of all roots (including dicot roots). In dicot stems, the endodermis is often rich in starch grains, which is why it is specifically referred to as the "starch sheath". Bulliform cells are large, empty, colorless cells found in the upper epidermis of grass leaves. In monocot stems, the hypodermis is made of sclerenchyma cells. In contrast, dicot stems have a collenchymatous hypodermis.

**(NCERT Ed. 2025-26, 11th, Page No. 74, 75, 76, 77)**

**98. (4)**

Non-living objects also grow if we take increase in body mass as a criterion for growth. Mountains, boulders and sand mounds do grow. However, this kind of growth exhibited by non-living objects is by accumulation of material on the surface. In living organisms, growth is from inside. Growth, therefore, cannot be taken as a defining property of living organisms. Growth is measured by a variety of parameters some of which are: increase in fresh weight, dry weight, length, area, volume and cell number. You may find it amazing to know that one single maize root apical meristem can give rise to more than 17,500 new cells per hour, whereas cells in a watermelon may increase in size by upto 3,50,000 times. In the former, growth is expressed as increase in cell number; the latter expresses growth as increase in size of the cell.

**(NCERT Ed. 2025-26, 11th, Page No. 168)**



[Contd...

**99. (2)**

The generative cell of a mature pollen grain divides mitotically to give rise to the two male gametes.

**(NCERT Ed. 2025-26, 12th, Page No. 7)**

**100. (4)**

Primary treatment : These treatment steps basically involve physical removal of particles – large and small – from the sewage through filtration and sedimentation. In the secondary treatment, biological decomposition of organic matter is done.

**(NCERT Ed. 2025-26, 12th, Page No. 154)**

**101. (1)**

Law of Segregation: This law is based on the fact that the alleles do not show any blending and that both the characters are recovered as such in the $F_2$ generation though one of these is not seen at the $F_1$ stage.

**(NCERT Ed. 2025-26, 12th, Page No. 59)**

**102. (3)**

The size of a population for any species is not a static parameter. It keeps changing with time. The size of a population depends on various factors including food availability, predation pressure and adverse weather.

**(NCERT Ed. 2025-26, 12th, Page No. 192)**

**103. (2)**

In a pond ecosystem, the autotrophic components include the phytoplankton, some algae and the floating, submerged and marginal plants found at the edges. The consumers are represented by the zooplankton, the free swimming and bottom dwelling forms.

**(NCERT Ed. 2025-26, 12th, Page No. 206)**

**104. (2)**

In general, species diversity decreases as we move away from the equator towards the poles.

**(NCERT Ed. 2025-26, 12th, Page No. 219, 220)**

**105. (3)**

Schleiden and Schwann together formulated the cell theory. This theory however, did not explain as to how new cells were formed. Rudolf Virchow (1855) first explained that cells divided and new cells are formed from pre-existing cells (*Omnis cellula-e cellula*). He modified the hypothesis of Schleiden and Schwann to give the cell theory a final shape.

**(NCERT Ed. 2025-26, 11th, Page No. 88)**

**106. (4)**

The plant body of a liverwort (bryophyte) is thalloid, e.g., *Marchantia*. Evolutionarily, the pteridophytes are the first terrestrial plants to possess vascular tissues – xylem and phloem.

**(NCERT Ed. 2025-26, 11th, Page No. 31)**

**107. (1)**

Man - Order Primata, Housefly - Order Diptera, Mango - Order Sapindales, Wheat - Order Poales

**(NCERT Ed. 2025-26, 11th, Page No. 8)**

**108. (3)**

The $C_3$ and $C_4$ plants respond differently to $CO_2$ concentrations. At low light conditions neither group responds to high $CO_2$ conditions. At high light intensities, both $C_3$ and $C_4$ plants show increase in the rates of photosynthesis. What is important to note is that the $C_4$ plants show saturation at about 360 $\mu l L^{-1}$ while $C_3$ responds to increased $CO_2$ concentration and saturation is seen only beyond 450 $\mu l L^{-1}$. Thus, current availability of $CO_2$ levels is limiting to the $C_3$ plants.

**(NCERT Ed. 2025-26, 11th, Page No. 150)**

**109. (2)**

In glycolysis, glucose undergoes partial oxidation to form two molecules of pyruvic acid.

**(NCERT Ed. 2025-26, 11th, Page No. 155)**

**110. (1)**

*Escherichia coli* has $4.6 \times 10^6$ bp, and haploid content of human DNA is $3.3 \times 10^9$ bp. A typical nucleosome contains 200 bp of DNA helix. The size of VNTR varies in size from 0.1 to 20 kb.

**(NCERT Ed. 2025-26, 12th, Page No. 80, 83, 106)**

**111. (2)**

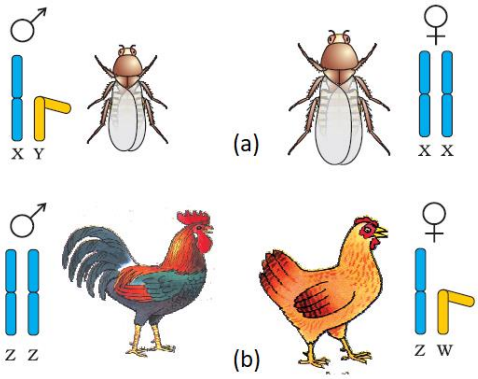
The outside of the epidermis is often covered with a waxy thick layer called the cuticle which prevents the loss of water. Cuticle is absent in roots.

**(NCERT Ed. 2025-26, 11th, Page No. 72)**



[Contd...

**112. (3)**




Determination of sex by chromosomal differences: (a) In *Drosophila*, the female has a pair of XX chromosomes (homogametic) and the male XY (heterogametic) composition; (b) In many birds, female has a pair of dissimilar chromosomes ZW and male two similar ZZ chromosomes

**(NCERT Ed. 2025-26, 12th, Page No. 70)**

**113. (3)**

Meiosis involves two sequential cycles of nuclear and cell division called meiosis I and meiosis II but only a single cycle of DNA replication.

**(NCERT Ed. 2025-26, 11th, Page No. 125)**

**114. (3)**

In general, viruses that infect plants have single stranded RNA and viruses that infect animals have either single or double stranded RNA or double stranded DNA. Bacterial viruses or bacteriophages (viruses that infect the bacteria) are usually double stranded DNA viruses. The protein coat called capsid made of small subunits called capsomeres, protects the nucleic acid.

**(NCERT Ed. 2025-26, 11th, Page No. 20)**

**115. (2)**

Self-incompatibility is a genetic mechanism and prevents self-pollen (from the same flower or other flowers of the same plant) from fertilising the ovules by inhibiting pollen germination or pollen tube growth in the pistil.

**(NCERT Ed. 2025-26, 12th, Page No. 15)**

**116. (2)**

The spores in pteridophytes germinate to give rise to inconspicuous, small but multicellular, free-living, mostly photosynthetic thalloid gametophytes called prothallus.

**(NCERT Ed. 2025-26, 11th, Page No. 32)**

**117. (4)**

In the floral formula, Br stands for bracteate K stands for calyx , C for corolla, P for perianth, A for androecium and G for Gynoecium, $\underline{G}$ for superior ovary and $\overline{G}$ for inferior ovary, ♂ for male, ♀ for female, ⚥ for bisexual plants, ⊕ for actinomorphic and % for zygomorphic nature of flower.

**(NCERT Ed. 2025-26, 11th, Page No. 67-68)**

**118. (2)**

| Stage | NADH Produced | ATP via Oxidative Phosphorylation |
|---|---|---|
| Glycolysis | 2 | 6 ATP |
| Link Reaction | 2 | 6 ATP |
| Krebs Cycle | 6 | 18 ATP |
| Total | 10 NADH | 30 ATP |

**(NCERT Ed. 2025-26, 11th, Page No. 160)**

**119. (1)**

Ernst Mayr, the Harvard University evolutionary biologist who has been called 'The Darwin of the 20th century', was one of the 100 greatest scientists of all time.

**(NCERT Ed. 2025-26, 11th, Page No. 2)**

**120. (3)**

Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents.

**(NCERT Ed. 2025-26, 12th, Page No. 153)**

**121. (2)**

Mitochondria are the sites of aerobic respiration. They produce cellular energy in the form of ATP, hence they are called 'power houses' of the cell. They divide by fission.

**(NCERT Ed. 2025-26, 11th, Page No. 97)**

**122. (4)**

Archaebacteria are special since they live in some of the most harsh habitats such as extreme salty areas (halophiles), hot springs (thermoacidophiles) and marshy areas (methanogens). Archaebacteria differ from other bacteria in having a different cell wall structure and this feature is responsible for their survival in extreme conditions. Methanogens are present in the gut of several ruminant animals such as cows and buffaloes and they are responsible for the production of methane (biogas) from the dung of these animals. These grow anaerobically on cellulosic material, produce large amount of methane along with $CO_2$ and $H_2$. These bacteria are collectively called methanogens, and one such common bacterium is *Methanobacterium*.

**(NCERT Ed. 2025-26, 11th, Page No. 13)**



[Contd...

**123. (1)**
Auxin controls xylem differentiation and helps in cell division.
**(NCERT Ed. 2025-26, 11th, Page No. 176)**

**124. (3)**
Gymnosperms have albuminous cells and sieve cells. They lack sieve tubes and companion cells. Sieve tube elements are also long, tube-like structures, arranged longitudinally and are associated with the companion cells. Their end walls are perforated in a sieve-like manner to form the sieve plates. A mature sieve element possesses a peripheral cytoplasm and a large vacuole but lacks a nucleus. The functions of sieve tubes are controlled by the nucleus of companion cells. The companion cells are specialised parenchymatous cells, which are closely associated with sieve tube elements.
**(NCERT Ed. 2022-23, 11th, Page No. 88)**

**125. (2)**
In PS I the reaction centre chlorophyll *a* has an absorption peak at 700 nm, hence is called P700.
**(NCERT Ed. 2025-26, 11th, Page No. 138)**

**126. (4)**
In a transverse section, a typical microsporangium appears near circular in outline. It is generally surrounded by four wall layers – the epidermis, endothecium, middle layers and the tapetum.
**(NCERT Ed. 2025-26, 12th, Page No. 5)**

**127. (4)**
Mother - $X^CX$, Father - XY
The son can inherit either $X^C$ or X allele from mother, thus having 50% chances of being color blind.
**(NCERT Ed. 2025-26, 12th, Page No. 73)**

**128. (3)**
There are at least three RNA polymerases in the nucleus (in addition to the RNA polymerase found in the organelles). There is a clear cut division of labour. The RNA polymerase I transcribes rRNAs (28S, 18S, and 5.8S), whereas the RNA polymerase III is responsible for transcription of tRNA, 5srRNA, and snRNAs (small nuclear RNAs). The RNA polymerase II transcribes precursor of mRNA, the heterogeneous nuclear RNA (hnRNA).
**(NCERT Ed. 2025-26, 12th, Page No. 95)**

**129. (3)**
Connell's elegant field experiments showed that on the rocky sea coasts of Scotland, the larger and competitively superior barnacle *Balanus* dominates the intertidal area, and excludes the smaller barnacle *Chathamalus* from that zone. This interaction is competition.
**(NCERT Ed. 2025-26, 12th, Page No. 199)**

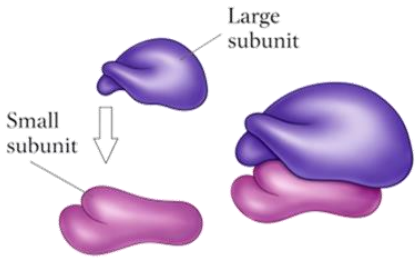
**130. (1)**
Trees occupy top vertical strata or layer of a forest, shrubs the second and herbs and grasses occupy the bottom layers.
**(NCERT Ed. 2025-26, 12th, Page No. 206)**

**131. (3)**
Habitat loss and fragmentation: This is the most important cause driving animals and plants to extinction. The Amazon rain forest (it is so huge that it is called the 'lungs of the planet') harbouring probably millions of species is being cut and cleared for cultivating soya beans or for conversion to grasslands for raising beef cattle.
**(NCERT Ed. 2025-26, 12th, Page No. 222)**

**132. (4)**
A classical example of point mutation is a change of single base pair in the gene for beta globin chain that results in the change of amino acid residue glutamate (GAG) to valine (GUG).
**(NCERT Ed. 2025-26, 12th, Page No. 96)**

**133. (2)**
The DNA-dependent DNA polymerases catalyse polymerisation only in one direction, that is 5'→3'.
**(NCERT Ed. 2025-26, 12th, Page No. 90)**

**134. (4)**
When the small subunit encounters an mRNA, the process of translation of the mRNA to protein begins. In prokaryotes, 'A' is 50S. In eukaryotes, 'B' is 40S.



Ribosome
**(NCERT Ed. 2025-26, 12th, Page No. 99)**

**135. (1)**
The food in red algae is stored as floridean starch which is very similar to amylopectin and glycogen in structure.
**(NCERT Ed. 2025-26, 11th, Page No. 27)**

**136. (2)**

Echinoderms possess a unique water vascular system which is a coelomic derivative (mesodermal). It helps in locomotion, food capture, and respiration. This is distinct from the water canal system of sponges (Porifera), which is formed by the body wall folding into a central spongocoel (not a true coelom).

**(NCERT Ed. 2025-26, 11th, Page No. 45)**

**137. (1)**

Proteins are often represented as a linear chain of amino acids, with the N-terminal (amino end) on the left and the C-terminal (carboxyl end) on the right when drawn schematically.

The tertiary structure gives the protein its specific three-dimensional shape, which is essential for biological activity, especially in enzymes and functional proteins.

In competitive inhibition, the Km increases, not decreases, because the substrate must compete with the inhibitor for the active site. Vmax remains unchanged.

Coenzymes are organic molecules that bind loosely and transiently to the enzyme. Molecules that are tightly and permanently bound are called prosthetic groups.

The enzyme Carbonic anhydrase can increase the rate of conversion of $CO_2 + H_2O \rightarrow H_2CO_3$ by about 10 million times.

In the absence of any enzyme this reaction is very slow, with about 200 molecules of H2 CO3 being formed in an hour. However, by using the enzyme present within the cytoplasm called carbonic anhydrase, the reaction speeds dramatically with about 600,000 molecules being formed every second.

**(NCERT Ed. 2025-26, 11th, Page No. 112)**

**138. (4)**

At the junction of foregut and midgut, a ring of 6–8 gastric caeca is present. These increase the surface area for digestion and secretion.

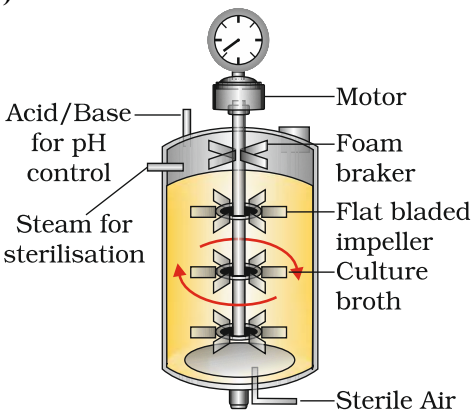
The hindgut is broader than the midgut and is differentiated into ileum, colon, and rectum.

Excretion occurs through 100–150 Malpighian tubules, which are yellow, thin, filamentous structures lined by glandular and ciliated cells.

The heart pumps haemolymph anteriorly, not posteriorly. Blood from sinuses enters the 13-chambered heart through ostia and is pumped towards the head (anterior direction) into the sinuses.

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**139. (2)**




A is foam breaker, B is motor, C is sterile air, and D is inlet for steam for sterilization.

**(NCERT Ed. 2025-26, 12th, Page No. 174)**

**140. (1)**

| List-I | List-II |
|---|---|
| P-wave | Depolarisation of the atria |
| Tunica intima | An inner lining of squamous endothelium |
| T-wave | Return of ventricles from excited to normal state |
| Tunica media | Comparatively thin in the veins |

**(NCERT Ed. 2025-26, 11th, Page No. 201)**

**141. (3)**

The hormone Aldosterone is secreted by the zona glomerulosa of the Adrenal cortex and is mainly regulated by the Renin–Angiotensin–Aldosterone System (RAAS).

When glomerular blood flow or blood pressure falls, the juxtaglomerular cells of the kidney release Renin. Renin converts angiotensinogen into angiotensin I, which is then converted to Angiotensin II. Angiotensin II stimulates the adrenal cortex to release aldosterone. Aldosterone increases $Na^+$ and water reabsorption, helping restore blood volume and pressure.

**(NCERT Ed. 2025-26, 11th, Page No. 212)**

**142. (2)**

Calcium ions bind to a specific subunit of troponin (not tropomyosin). This binding causes a conformational change that pulls tropomyosin away from the active sites on the actin filament, allowing the myosin head to bind and initiate contraction.

**(NCERT Ed. 2025-26, 11th, Page No. 222)**

**143. (2)**

The sodium-potassium pump actively transports 3 $Na^+$ outwards for 2 $K^+$ into the cell, maintaining the concentration gradient against leak channels. If the pump is blocked, ions will slowly move down their concentration gradients through passive leak channels, and the gradient will gradually be lost.

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[Contd...

**144. (3)**

| List-I | List-II |
|---|---|
| Glucagon | Acts as antagonist to insulin |
| Cortisol | A hypocalcemic hormone |
| Aldosterone | Mineralocorticoid |
| Thyrocalcitonin | Stimulates lipolysis, and proteolysis |

**(NCERT Ed. 2025-26, 11th, Page No. 245)**

**145. (1)**

In human females, oogenesis begins during embryonic development. The primordial germ cells differentiate into oogonia, which enter meiosis and become primary oocytes before birth. These primary oocytes remain arrested in prophase-I until puberty.

In males, spermatogenesis starts at puberty. At this stage, the hypothalamus increases secretion of Gonadotropin-Releasing Hormone (GnRH), which stimulates the anterior pituitary to release FSH and LH. These hormones activate the testes and initiate spermatogenesis.

**(NCERT Ed. 2025-26, 12th, Page No. 31)**

**146. (2)**

LNG-20 is a hormone-releasing intrauterine device (IUD) that releases Levonorgestrel.

Hormone-releasing IUDs:

- Make the endometrium unsuitable for implantation, and
- Thicken cervical mucus, making the cervix hostile to sperms.

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**147. (4)**

Let the recessive allele be q and the dominant allele be p.

Frequency of recessive phenotype = $q^2 = 0.16$

$q = \sqrt{0.16} = 0.4$

Since $p + q = 1$:

$p = 1 - q = 1 - 0.4 = 0.6$

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**148. (1)**

Only pair III is incorrectly matched. Malignant malaria, which is the most serious and potentially fatal form, is caused by *Plasmodium falciparum*, not *Plasmodium vivax*. *Wuchereria bancrofti* and W. malayi cause elephantiasis. Ringworm is caused by Microsporum, Trichophyton, and Epidermophyton.

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**149. (3)**

The restriction enzyme *EcoR* I recognizes the sequence GAATTC and cuts between G and A, producing sticky ends with 4-nucleotide overhangs (AATT).

The alien circular DNA has 3 *EcoR* I sites, so digestion produces 3 fragments.

The vector pBR322 has one *EcoR* I site, so it becomes linear with two sticky ends.

To insert one fragment, two junctions are formed between the vector and insert.

Each *EcoR* I sticky end:

- Overhang = AATT
- Base pairing: A–T (2 H-bonds)

So hydrogen bonds per sticky end:

4 base pairs × 2=8 H-bonds

Two junctions are formed:

$8 \times 2 = 16$

Thus 16 hydrogen bonds must form during annealing before ligation by DNA ligase to integrate the fragment.

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**150. (3)**

If the functional ADA gene is introduced at the early embryonic stage, it can integrate into cells that will divide and give rise to many tissues. This can potentially provide a permanent cure for ADA Deficiency.

In the gene therapy method described, lymphocytes are removed, cultured, and infected with a retroviral vector carrying ADA cDNA, then reintroduced into the patient. However, lymphocytes are not immortal cells; they have a finite lifespan. Therefore, the therapy must be repeated periodically.

**(NCERT Ed. 2025-26, 12th, Page No. 182)**

**151. (3)**

Arthropoda, the largest phylum of Animalia, is uniquely characterized by a chitinous exoskeleton, jointed appendages (arthros-joint, poda-appendages), and excretion via Malpighian tubules (especially in insects).

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[Contd...

**152. (1)**

At the Isoelectric Point (pI) of an amino acid, the molecule exists predominantly as a Zwitterion, where:

- The amino group ($–NH_3^+$) carries a positive charge
- The carboxyl group ($–COO^–$) carries a negative charge

These opposite charges balance each other, so the net charge on the amino acid is zero (electrically neutral).

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**153. (2)**

Gap junctions allow neighbouring cells to communicate by transferring ions and small molecules.

Adhering junctions (adherens junctions) act like cementing structures, keeping adjacent cells together.

Compound epithelium consists of multiple layers of cells, not a single layer. A single layer of tall slender cells describes simple columnar epithelium.

In Connective Tissue, cells secrete fibres such as collagen and elastin, except in blood, where fibres are absent in normal conditions.

Smooth Muscle fibres are fusiform, non-striated and involuntary.

**(NCERT Ed. 2020-21, 11th, Page No. 102)**

**154. (3)**

Breathing is regulated by the Respiratory Rhythm Centre located in the Medulla Oblongata.

Two types of chemoreceptors detect changes in $CO_2$ and $H^+$ concentration:

1. Central chemoreceptors:

The Chemosensitive Area near the respiratory rhythm centre senses increased $H^+$ concentration (due to $CO_2$) in cerebrospinal fluid and signals the rhythm centre to adjust breathing.

2. Peripheral chemoreceptors:

Receptors in the Carotid Body and Aortic Body detect changes in $CO_2$, $H^+$, and $O_2$ levels in blood and send signals to the respiratory centre.

**(NCERT Ed. 2025-26, 11th, Page No. 190)**

**155. (1)**

Cardiac output is calculated using:

Cardiac Output = Stroke Volume × Heart Rate

Stroke Volume (SV) = End Diastolic Volume − End Systolic Volume

- End Diastolic Volume (EDV) = 120 mL
- End Systolic Volume (ESV) = 50 mL

SV = 120 – 50 = 70 mL

Heart rate = 75 beats/min

Cardiac Output = 70 × 75

= 5250 mL/min

This represents the volume of blood pumped by the heart per minute by the Human Heart.

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**156. (3)**

In the Loop of Henle, filtrate flows in opposite directions in the descending and ascending limbs, forming part of the Counter-current Mechanism.

The descending limb is permeable to water but almost impermeable to electrolytes (NaCl).

The ascending limb actively transports NaCl, which contributes to the osmotic gradient and interacts with the Vasa Recta through counter-current exchange.

Small amounts of urea enter the thin segment of the ascending limb, contributing to medullary osmotic gradient.

Human kidneys can produce urine about four times more concentrated than the initial filtrate.

**(NCERT Ed. 2025-26, 11th, Page No. 209)**

**157. (3)**

The 8th, 9th, and 10th pairs of ribs are called vertebrochondral (false) ribs because they do not articulate directly with the sternum but attach to the seventh rib via hyaline cartilage. The 11th and 12th pairs are floating ribs.

**(NCERT Ed. 2025-26, 11th, Page No. 225)**

**158. (3)**

The midbrain lies between the forebrain (specifically the thalamus and hypothalamus of the diencephalon) and the pons of the hindbrain.

The brain stem is not a part of the spinal cord.

The brain stem is a part of the brain and consists of:

- Midbrain
- Pons
- Medulla oblongata

The medulla oblongata continues into the spinal cord, but the brain stem itself belongs to the brain, not the spinal cord.

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[Contd...

**159. (2)**

Hormones that interact with intracellular (cytoplasmic or nuclear) receptors are typically lipid-soluble hormones. These hormones can diffuse through the plasma membrane and bind to receptors inside the cell, forming a hormone–receptor complex that regulates gene expression on DNA.

Steroid hormones belong to this category.

Cortisol, Testosterone, Estradiol

These hormones bind intracellular receptors and directly influence transcription and chromosome function.

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**160. (1)**

Menstruation occurs due to the breakdown of the endometrial lining, triggered by a drop in progesterone when the corpus luteum degenerates. If fertilization occurs (pregnancy), hCG maintains the corpus luteum, which continues to secrete progesterone. This maintains the endometrium, preventing menstruation.

**(NCERT Ed. 2025-26, 12th, Page No. 35)**

**161. (2)**

The 2017 amendment states that a pregnancy can be terminated within the first 12 weeks on the opinion of one registered medical practitioner. If the pregnancy has lasted more than 12 weeks but fewer than 24 weeks, the opinion of two registered medical practitioners is required.

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**162. (3)**

*Homo erectus* lived about 1.5 million years ago. Fossils show a brain capacity around 900 cc.

Neanderthal man had a brain size of about 1400 cc. However, they did not live in East African grasslands. They lived in near-east and central Asia and Europe about 100,000–40,000 years ago.

**(NCERT Ed. 2025-26, 12th, Page No. 125)**

**163. (3)**

Diseases in which the immune system attacks the body's own cells are called autoimmune diseases.

Myasthenia gravis is an autoimmune disorder where antibodies attack acetylcholine receptors at the neuromuscular junction, causing muscle weakness and fatigue.

Gout – caused by uric acid crystal deposition in joints, not an autoimmune attack.

Asthma – a hypersensitivity/allergic reaction, not a true autoimmune disease.

Osteoporosis – loss of bone density due to hormonal and metabolic factors.

**(NCERT Ed. 2025-26, 11th, Page No. 227)**

**164. (2)**

| List-I | List-II |
|---|---|
| Lysozyme | Degradation of bacterial cell wall |
| Genetic engineering | Techniques to alter the chemistry of genetic material |
| Chitinase | Degradation of fungal cell wall |
| Plasmid | Autonomously replicating circular extra-chromosomal DNA |

**(NCERT Ed. 2025-26, 12th, Page No. 171)**

**165. (2)**

Protection of tobacco plants against the nematode *Meloidogyne incognita* was achieved using RNA interference (RNAi).

In this method:

- A gene from the nematode was introduced into the Tobacco plant using a vector.
- The inserted DNA produced both sense RNA and antisense RNA.
- These RNAs paired to form double-stranded RNA (dsRNA).
- The dsRNA triggered RNA interference, which silenced the nematode's specific mRNA.
- As a result, the nematode could not survive in the plant, providing resistance.

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**166. (3)**

Members of Osteichthyes (bony fishes) show the following characteristics:

- They generally possess four pairs of gills covered by an operculum.
- Their skin is covered with cycloid or ctenoid scales.

Poison sting or electric organs are characteristic features of Chondrichthyes (cartilaginous fishes) such as *Trygon* and *Torpedo*.

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**167. (1)**

In a polysaccharide chain like glycogen, the right end is called the reducing end and the left end is called the non-reducing end. N-terminal and C-terminal refer to proteins.

**(NCERT Ed. 2025-26, 11th, Page No. 110)**

**168. (3)**

Male frogs can be externally distinguished from female frogs by the presence of vocal sacs and a copulatory pad specifically on the first digit of the forelimbs.

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[Contd...

**169. (1)**

Functional Residual Capacity (FRC) is the volume of air that remains in the lungs after a normal expiration. It includes the Expiratory Reserve Volume (ERV) and the Residual Volume (RV). FRC = ERV + RV = 1100 mL + 1200 mL = 2300 mL.

**(NCERT Ed. 2025-26, 11th, Page No. 187)**

**170. (3)**

Pairs I, II, and III are correctly matched. Pair IV is incorrect because B-lymphocytes are responsible for humoral (antibody-mediated) immunity, whereas T-lymphocytes are responsible for cell-mediated immunity.

**(NCERT Ed. 2025-26, 11th, Page No. 194)**

**171. (4)**

| List-I | List-II |
|---|---|
| Hyoid | U-shaped bone |
| Joints | Acts as fulcrum |
| Addison's disease | Hyposecretion of hormones of adrenal cortex |
| Collecting Duct | Allows passage of small amounts of urea |

**(NCERT Ed. 2025-26, 11th, Page No. 244)**

**172. (1)**

In frogs, liver secretes bile that is stored in the gall bladder.

Cardiac muscle cells possess Intercalated discs, which help transmit impulses rapidly so the heart contracts as a coordinated unit.

Frogs are beneficial for mankind because they eat insects and protect the crop. Frogs maintain ecological balance because these serve as an important link of food chain and food web in the ecosystem. In some countries the muscular legs of frog are used as food by man.

The A-band contains thick myosin filaments and overlapping thin actin filaments, not only thin filaments.

**(NCERT Ed. 2025-26, 11th, Page No. 221)**

**173. (3)**

The limbic system along with the hypothalamus regulates sexual behaviour, emotional reactions, and motivation. Cardiovascular reflexes and gastric secretions are primarily controlled by centers located in the medulla oblongata of the hindbrain.

**(NCERT Ed. 2025-26, 11th, Page No. 236)**

**174. (4)**

Convergent evolution occurs when unrelated species independently evolve similar traits to adapt to similar environmental pressures, even though their ancestry is different.

Forelimbs of whales, bats, cheetahs, and humans are homologous structures (same origin, different functions).

Vertebrate hearts and brains are homologous (common ancestry).

Thorns of *Bougainvillea* and tendrils of *Cucurbita* are analogous but less commonly cited.

Eyes of octopus and mammals are analogous organs, structurally different in development but functionally similar, making them a classic example of convergent evolution.

**(NCERT Ed. 2025-26, 12th, Page No. 115)**

**175. (2)**

During recombinant DNA isolation, after removing proteins and RNA, DNA is precipitated by adding chilled ethanol or isopropanol.

DNA is insoluble in alcohol, causing it to come out of solution as a visible precipitate.

Ribonuclease → removes RNA

Ethidium bromide → used for staining DNA in gels

Agarose gel → medium for gel electrophoresis

**(NCERT Ed. 2025-26, 12th, Page No. 171)**

**176. (3)**

In *In-Vitro* Fertilization (IVF) programs, the timing and location of the embryo transfer depend on its stage of development:

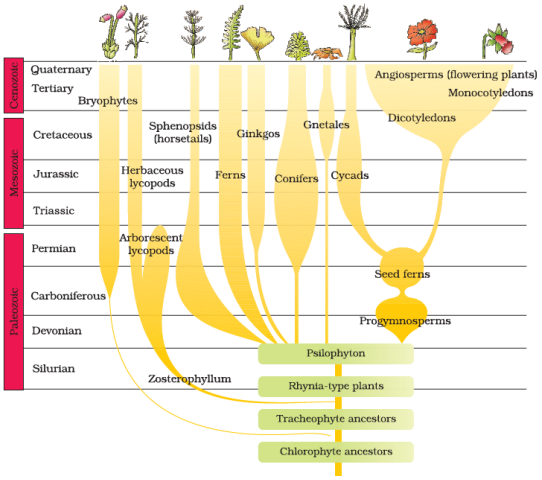
- ZIFT (Zygote Intrafallopian Transfer): If the embryo is at the zygote stage or an early embryo stage with up to 8 blastomeres, it is transferred into the fallopian tube.
- IUT (Intrauterine Transfer): If the embryo has developed beyond the 8-blastomere stage (into a late morula or blastocyst), it is transferred directly into the uterus to complete its further development.

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[Contd...

**177. (1)**




The transition from aquatic ancestors to terrestrial flowering plants followed this general progression:

1. Chlorophyte ancestors: Green algae that represent the earliest ancestors of all land plants.
2. Tracheophyte ancestors: The emergence of vascular tissues (xylem and phloem).
3. Rhynia-type plants: Early, simple vascular plants that lacked true leaves and roots.
4. Psilophyton: A more complex ancestral vascular plant that served as a common progenitor for several lineages.
5. Seed ferns (Pteridosperms): The critical transitional group that developed the ability to produce seeds, eventually leading to both Gymnosperms and Angiosperms.
6. Angiosperms: The modern flowering plants.

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**178. (3)**

Benign tumors remain localised, grow slowly, and do not invade surrounding tissues.

Malignant tumors do not rely on contact inhibition; in fact, loss of contact inhibition is a hallmark of malignant cells, which allows them to grow uncontrollably and invade other tissues.

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**179. (4)**

In plasmids like pBR322, the *BamH* I site is located within the tetracycline resistance ($tet^R$) gene, so insertion at this site causes insertional inactivation of $tet^R$.

The recombinant plasmid does not lose ampicillin resistance ($amp^R$); only the tetracycline resistance is affected. $amp^R$ is controlled by a different gene not interrupted by *BamH* I insertion.

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**180. (2)**

In 1997, the first transgenic cow, Rosie, produced human protein-enriched milk containing human alpha-lactalbumin at a concentration of 2.4 grams per litre.

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**PW Web/App** - https://smart.link/7wwosivoicgd4

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